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Revisiting the age and paleomagnetism of the Modipe Gabbro of South Africa

S.W. Denyszyn[a,*], J.M. Feinberg[b], P.R. Renne[c,d], G.R. Scott[c]

[a] School of Earth and Environment/Centre for Exploration Targeting, University of Western Australia, 35 Stirling Highway, Crawley, WA 6009, Australia
[b] Institute for Rock Magnetism, Department of Earth Sciences, University of Minnesota, 100 Union Street SE, Minneapolis, MN 55455, USA
[c] Berkeley Geochronology Center, 2455 Ridge Road, Berkeley, CA 94709, USA
[d] Department of Earth and Planetary Science, University of California, Berkeley, CA 94720, USA

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ABSTRACT

The Modipe Gabbro is a Precambrian intrusive unit hosted by the Kaapvaal craton of South Africa and provides one of the oldest records of geomagnetic field behavior in the late Archean. Earlier paleomagnetic research into the Modipe Gabbro provided valuable insight into its magnetic properties and mineralogy, but predate the development of paleomagnetic tools that are important in unraveling potentially complex histories of Precambrian rocks. Further, a lack of information about the precise age of the intrusion prevented this earlier paleomagnetic study from being used to evaluate the tectonic history of the Kaapvaal craton. Here we present a comprehensive paleomagnetic and geochronologic study of the Modipe Gabbro that complements and builds upon the existing work. The emplacement age of the gabbro (2784.0 ± 1.0 Ma), established using the U–Pb ID-TIMS method on baddeleyite, links the intrusion to nearby igneous bodies in the Kaapvaal craton, establishing one of the oldest fully differentiated magmatic events in the geologic record. The precise age of the Modipe Gabbro and field observations about its strike allow us to perform a tilt correction using the both the attitude and paleomagnetism of the nearby Derdepoort basalts. The tilt-corrected data allow a comparison of Kaapvaal craton's paleolatitude with that of the Pilbara craton at 2.78 Ga, with implications for the existence of the supercraton Vaalbara.
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1. Introduction

The transition from the Archean to the Paleoproterozoic (3.0–2.5 Ga) is a key period in Earth's geophysical and geochemical evolution. In particular, this period encompasses an episode of substantial continental crustal growth and preservation, thought to be related to periodic mantle overturn events and extensive plume volcanism (Condie, 2001). This reconstitution of the upper mantle by deeper hotter mantle was followed by, or was synchronous with, the assembly and breakup of some of the planet's earliest supercontinents or supercratons (Nance et al., 1986; Williams et al., 1991; Young et al., 2001; Cheney, 1996; Aspler and Chiarenzelli, 1998; Condie, 2001; Barley et al., 2005), and the development of oxygenated conditions in the atmosphere. Establishing links between these major environmental changes and the suspected mantle-plume tectonic regime that characterized the late Archean is an ongoing research focus that is often hampered by the absence of reliable paleomagnetic and geochronologic constraints.

One of the earliest and most influential paleomagnetic studies on Archean rocks was that of Evans and McElhinny (1966), who examined the Modipe Gabbro in the Kaapvaal craton. This intrusive unit outcrops as a chain of low rising hills, or inselbergs, approximately 15 km long extending from southern Botswana into northern South Africa (Fig. 1). The Modipe Gabbro was emplaced as a sub-horizontal, sill-like body and the low-lying areas between the gabbro inselbergs consist of soils derived from the Gaborone Granite (McElhinny, 1966). Although exposures between the gabbro and the granite are exceedingly rare, Evans and McElhinny (1966) reported a sheared contact between the two lithologies that suggested that the gabbro was intruded by the granite. Importantly, they reported a positive baked contact test between the two lithologies. They also noted exceptionally stable magnetizations in the gabbro with high coercivities (≥180 mT) and high unblocking temperatures (>500 °C) and identified elongate magnetite inclusions exsolved within clinopyroxene as the main carrier of remanence (Evans et al., 1968). Subsequent work on similar occurrences of magnetite exsolved from clinopyroxenes has shown that they are robust recorders of the geomagnetic field (e.g., Renne et al., 2002; Feinberg et al., 2005; Muxworthy et al., 2013). However, these exsolved magnetite crystals are not ubiquitous throughout the Modipe Gabbro, and identical, stable remanences are found in those

* Corresponding author. Tel.: +61 (8) 6488 7329.
E-mail address: steven.denyszyn@uwa.edu.au (S.W. Denyszyn).

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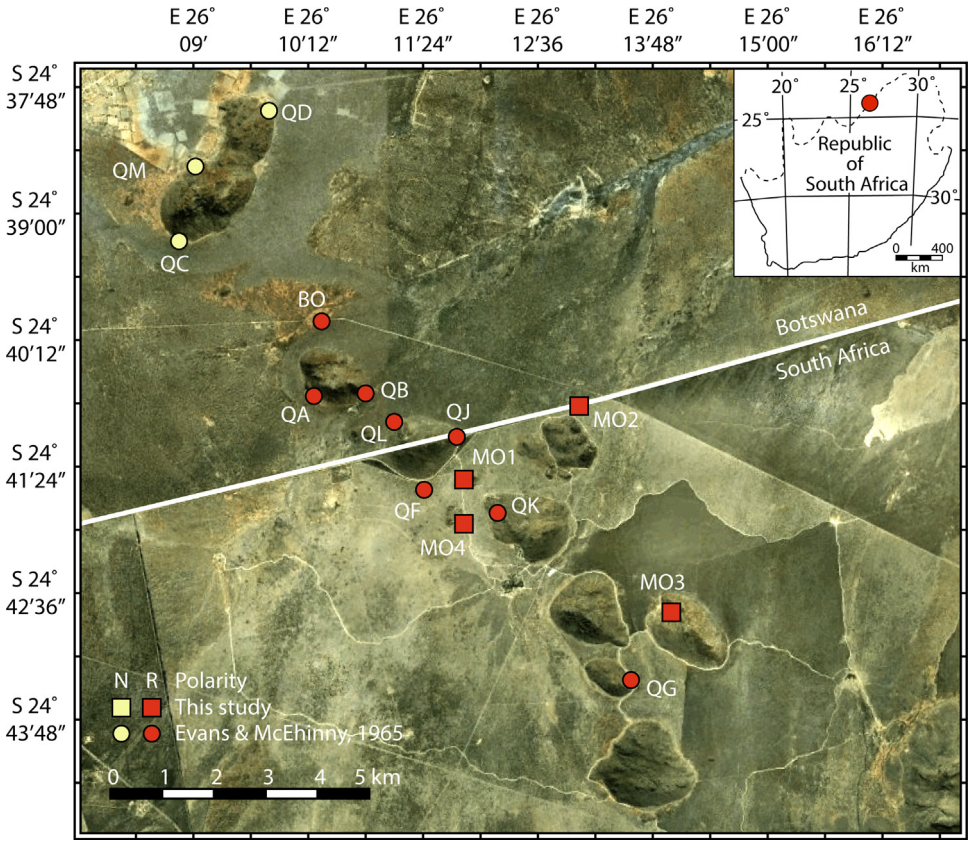


Fig. 1. Aerial view of the Modipe Complex across Botswana and South Africa. The Modipe Gabbro outcrops as low hills that rise above the otherwise flat topography of northern South Africa. Squares indicate sampling sites in this study, while circles show the sampling sites of Evans and McElhinny (1966). Red (yellow) symbols indicate normal (reversed) polarity. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

parts of the intrusion that do not contain them. There is, therefore, at least one other primary magnetic phase present, and we have identified it as part of this study. There are no clear paleohorizontal indicators within the Modipe Gabbro, which precludes evaluating whether a tilt-correction to paleomagnetic directions is needed. Evans and McElhinny (1966) established a paleolatitude of 80° for the Modipe Gabbro without using any form of tilt correction, and the corresponding virtual geomagnetic pole (VGP) calculated from the mean of their sampling sites (33.3° S and 31.1° E) is still used when discussing the tectonic history of the Kaapvaal craton. The exact position of the Kaapvaal craton relative to that of the Pilbara craton is especially important when considering the configurational history of the supercontinent Vaalbara.

Two years after their paleomagnetic analysis of the Modipe Gabbro, McElhinny and Evans (1968) estimated the strength of the Earth's ancient geomagnetic field, or paleointensity, at the time the unit was emplaced. The Earth's magnetic field is generated by convection of the liquid iron outer core and recordings of the strength of the field through time have been highly sought after to address questions about the stand-off distance of the solar wind (Tarduno et al., 2010) and the timing of the nucleation of solid inner core (Smirnov et al., 2003). This study has been updated by the recent work of Muxworthy et al. (2013) to indicate that magnetic field strength at the time of emplacement was approximately 80% of the modern field. However, the usefulness of this paleointensity estimate has been blunted due to ambiguity surrounding the Modipe Gabbro's precise paleolatitude and age.

Until now, the age of the Modipe Gabbro has been roughly constrained by Rb–Sr and K–Ar ages determined by McElhinny (1966). Rb–Sr total rock measurements yielded an age of 2630 ± 470 Ma. K–Ar measurements of pyroxene separates from three sites yielded ages ranging from ~2600 to ~3000 Ma with uncertainties of ±100 Ma. The average of these pyroxene K–Ar dates is 2758 ± 126 Ma when recalculated with the calibration of Renne et al. (2011) (Supplemental Data File). K–Ar measurements on plagioclase separates yielded anomalously young ages of 1983–1934 Ma, which were interpreted to possibly reflect thermal resetting by the nearby Bushveld Complex. The average K–Ar plagioclase age of McElhinny (1966), recalculated per Renne et al. (2011), is 1961 ± 52 Ma, which overlaps with integrated $^{40}Ar/^{39}Ar$ cooling ages for plagioclase from a Bushveld gabbro reported by Cassata et al. (2009). Thus, while portions of the paleomagnetic analysis of the Modipe Gabbro have long been known to be of high quality, it has been difficult to make use of these analyses owing to the poor geochronologic constraints on the unit.

Here we report the results of a new paleomagnetic and geochronologic study of the Modipe Gabbro. We make use of modern paleomagnetic protocols, including least-squares fitting and great circle analyses, to examine and refine the earlier paleomagnetic work of Evans and McElhinny (1966). Rock magnetic techniques are used to better characterize the extent of mineralogic alteration experienced by the magnetic oxides within the gabbro since their formation in the late Archean. The precise emplacement age of Modipe Gabbro is determined using U–Pb TIMS geochronological methods using baddeleyite as a geochronometer. These improved paleomagnetic and geochronologic data allow us to better constrain the precise position of the Kaapvaal craton at the time of the Modipe Gabbro's emplacement and to use the data from the Modipe Gabbro in reconstructions and evaluations of the supercontinent Vaalbara.

2. Petrography of the Modipe Gabbro

The Modipe Gabbro consists primarily of medium-grained labradoritic plagioclase and clinopyroxene, both of which exhibit progressively increasing levels of alteration toward the southeast. This spatial trend in degree of alteration, combined with the K–Ar plagioclase ages, is consistent with thermal effects of the Bushveld Complex, which outcrops 20 km to the south of the Modipe Complex. In more heavily altered samples, the clinopyroxene is replaced by actinolite and chlorite with minor amounts of secondary magnetite, while plagioclase is altered to sericite (Fig. 2). Plagioclase grains also exhibit low-level deformation features including bent twin planes, growth steps in twin boundaries, and deformation twins nucleating at high stress sites at grain boundaries (Fig. 2 and Supplemental Data File). This low-level alteration is similar to that observed in mafic dykes and sills that have undergone alteration by metasomatic fluids during and immediately after emplacement (e.g., France et al., 2009). The textures observed in the Modipe

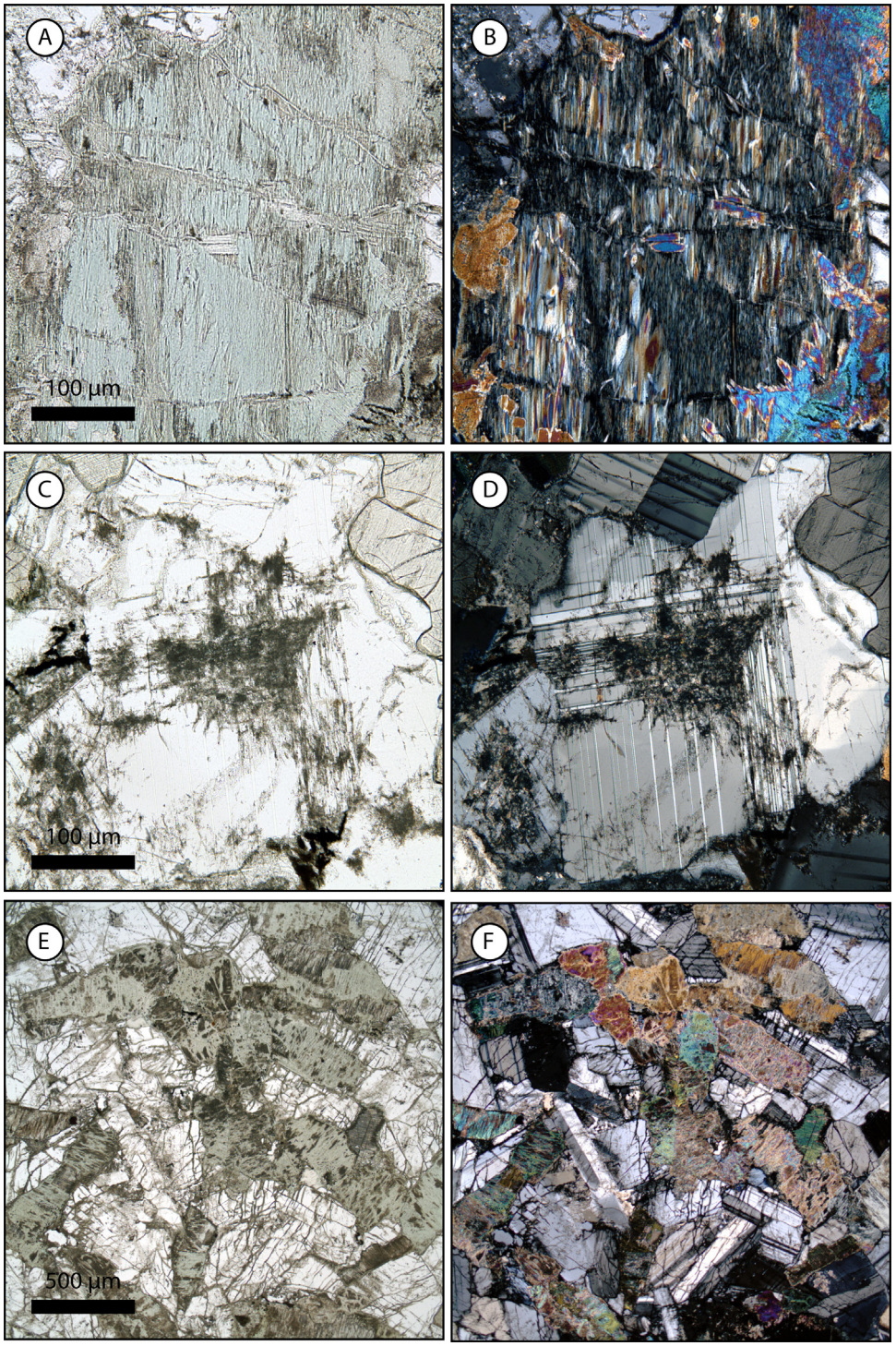


Fig. 2. Mineral alteration in the Modipe Gabbro. Replacement of pyroxene by actinolite and magnetite at site MO1-09, (A) plane polarized light, (B) cross-polarized light. Sericitization of plagioclase at site MO2-07, (C) PPL, (D) XPL, alteration consumed the original pyroxene, altered its inclusions, and produced a minor amount of additional magnetite at site MO4-01.

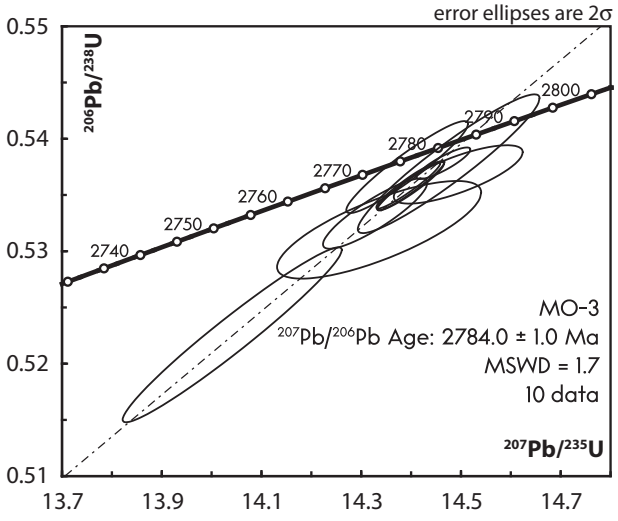


Fig. 3. U–Pb concordia diagram for sample MO-3. Ages on the concordia curve are in Ma. Errors on the ellipses and on the calculated age are 2σ.

pyroxenes are similar to those examined by Veblen and Buseck (1981) in which hydrated pyroxenes were shown to alter according to a multistep reaction involving (1) exsolution of amphibole lamellae, followed by (2) mass conversion to amphibole, triple-chain silicates, and pyriboles, and finally (3) conversion to sheet silicates such as mica, chlorite, and talc. This low-level metasomatic alteration likely occurred either during original emplacement of the Modipe Gabbro or during the subsequent emplacement of the Bushveld Complex toward the south. Despite this minimal alteration, the Modipe is exceptionally well preserved and the more northerly inselbergs show no petrographic evidence of regional metamorphism.

Fe-oxides observed in thin section include large (>300 μm) grains of primary magnetite decorating the edges of plagioclase laths, and secondary magnetite associated with the alteration of Fe–Mg silicate minerals such as pyroxene and olivine. No Fe-sulfides were observed. Thorough examination of thin sections from all four collection sites revealed no evidence of exsolved magnetite inclusions in pyroxene. Evans and McElhinny (1966) reported such inclusions in samples from just one site (BO), which is located on the Botswana side of the Modipe Gabbro. The samples in the present study were collected solely from the South African side of the Modipe Gabbro, suggesting that conditions in this portion of the Modipe were either (1) not suitable for the exsolution of magnetite from pyroxene, or (2) that the inclusions were sufficiently disturbed during alteration of the pyroxene that they have become obscured or destroyed.

3. U-Pb Geochronology

A description of the U–Pb isotopic methodology and results for sample MO-3 are given in the Supplemental Data File and the Concordia diagram of Fig. 3. Ten single baddeleyite crystals yield a $^{207}Pb/^{206}Pb$ age of 2784.0 ± 1.0 Ma (2σ), with an MSWD of 1.7. The uncertainty given does not include decay-constant errors; including them produces an uncertainty of ±7.0 Ma. Using the updated ^{235}U decay constant and uncertainty of Mattinson (2010), the $^{207}Pb/^{206}Pb$ age is 2779.9 ± 1.0 Ma without including decay-constant uncertainties, and ±2.7 Ma when they are included. For the purposes of consistency with previously published U–Pb ages discussed here, we report the age and uncertainty using only the decay constants of Jaffey et al. (1971) and ignoring the decay-constant errors. These errors must be considered however, in comparing our results with those from any other radioisotopic system. Th/U ratios are between 0.02 and 0.05, typical values for baddeleyite. Uranium concentrations are between 323 ppm and 770 ppm. The data are concordant or slightly discordant (up to 2.6% discordant, sample #10) (Fig. 3, Supplementary Table 2), but give consistently overlapping $^{207}Pb/^{206}Pb$ ages. The discordance is attributed to minor Pb loss, and the lower intercept of a regression line through all data goes through the origin, making any event that would disturb the U–Pb systematics of the baddeleyite grains unlikely.

In the context of establishing a precise and accurate age for the larger magmatic event using the most up-to-date radioisotopic parameters as of the time of this writing, the age of 2779.9 ± 1.0 [2.7] Ma must be the preferred age of emplacement. However, the geological interpretations of our data require direct comparison with previous work, which has not been re-calibrated using the new ^{235}U decay constant. Therefore, for our purposes the average $^{207}Pb/^{206}Pb$ age of 2784.0 ± 1.0 Ma reported here is considered the age of emplacement of the Modipe Gabbro.

4. Rock magnetism

Expanded descriptions of the rock magnetic and paleomagnetic methodologies are included in the supplemental information file.

4.1. Room temperature measurements

Mass normalized, in-phase susceptibility values range from 1.1×10^{-7} to 2.1×10^{-6} m^3 kg^{-1}, with a median value of 2.5×10^{-7} m^3 kg^{-1}. Susceptibilities from samples collected at the southernmost site, MO3, are three times greater than those of the other sites (Supplemental Data File). Other room-temperature magnetic properties from MO3 stand out when comparing results between sites. The median coercivity (H_c) at MO3 (8 mT) is significantly lower than that of MO1 (13 mT), MO2 (17 mT), or MO4 (15 mT). Similarly, the median "squareness" ratio, M_r/M_s, at MO3 (0.09) is notably lower than that of MO1 (0.13), MO2 (0.19), or MO4 (0.21).

4.2. Magnetic properties as a function of temperature

To determine the composition of the magnetic minerals in the Modipe Gabbro a subset of samples were selected for high-temperature susceptibility and strong-field thermomagnetic measurements. Measurements of susceptibility as a function of temperature revealed a mean Curie temperature of 579 ± 11 °C ($N = 11$), and the heating portion of the $\chi(T)$ experiments are shown in Fig. 4. Cooling curves are generally reversible and are not shown here in order to allow the reader to better evaluate changes in susceptibility during heating. The $\chi(T)$ experiments show that magnetic mineral concentrations vary across the Modipe Gabbro, and in some cases, ferromagnetic minerals are sufficiently rare that their signal in the $\chi(T)$ data is nearly obscured by the paramagnetic behavior of the silicate matrix minerals (e.g., amphibole, pyroxene, and olivine). Regardless of the concentration of magnetic material, the Curie temperature estimates are well clustered near 580 °C, the temperature expected for stoichiometric magnetite.

Strong-field thermomagnetic measurements produced a mean Curie temperature of 582 ± 6 °C (Fig. 4), in good agreement with the susceptibility data. These results are slightly lower than the 600 °C unblocking temperature reported in Evans and McElhinny (1966) and Evans et al. (1968), but are still within acceptable agreement, as these older estimates were determined using relatively coarse 6–8 step thermal demagnetization protocols.

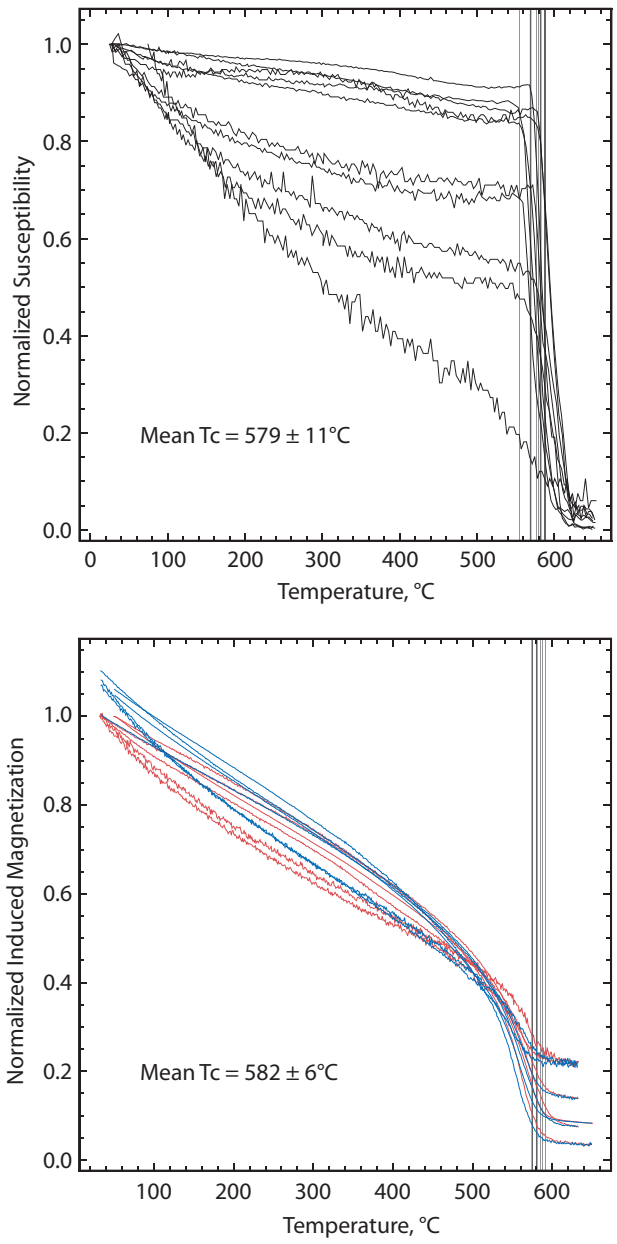


Fig. 4. Upper: in-phase magnetic susceptibility as a function of temperature ($n = 10$). Lower: strong-field thermomagnetic curves ($n = 6$). Red (blue) curves show heating (cooling) measurements. Vertical lines show Tc estimates. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

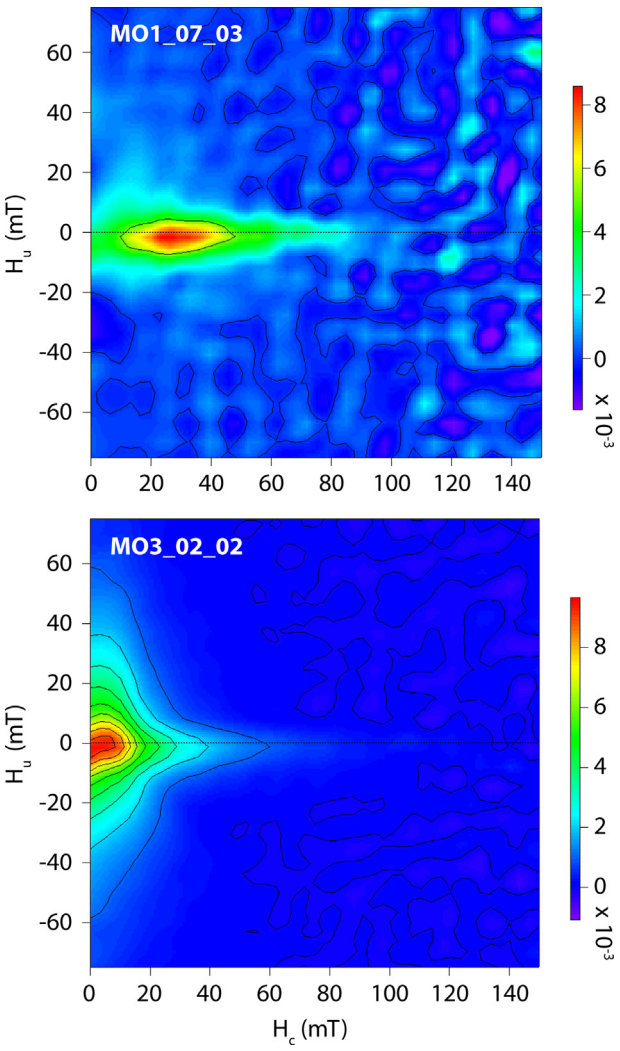


Fig. 5. Representative FORC diagrams for specimens that span the range of behavior on the squareness–coercivity plot. Both samples show a population of non-interacting single domain grains whose coercivities extend up to ~90 mT. MO3_02_02 is representative of more mineralogically altered specimens and contains a population of highly interacting, low coercivity grains. Both plots were processed using FORCinel with a smoothing factor of 2.6 (Harrison and Feinberg, 2008).

4.3. Hysteresis properties

Major hysteresis loops were collected for most specimens and the results of these measurements are summarized in a squareness–coercivity plot in the Supplemental Data File. Generally, the samples exhibit multidomain behavior with low coercivities and squareness values. However, first-order reversal diagrams of select samples reveal that there is actually a wide range of magnetic domain states within each sample. Fig. 5 shows representative FORC diagrams for samples that span the full range of behavior in the squareness–coercivity plot in the Supplemental Data File. Both samples contain a population of non-interacting, stable single domain grains whose median coercivity is ~25 mT with maximum values near 90 mT, and specimen MO1_07_03 typifies this distribution. In contrast, specimens that display significant mineral alteration, such as MO3_02_02, contain an additional population of highly interacting, low-coercivity grains. The thermomagnetic measurements described above suggest that the gabbro contains a single magnetic mineral carrier with a composition of nearly pure magnetite. Thus, we conclude that mineral alteration is responsible for the low coercivity magnetite component, as well as the elevated room temperature susceptibilities.

4.4. Magnetic unmixing

The low coercivity component observed in the FORC diagrams also appears as a distinct coercivity population when the IRM and ARM acquisition measurements are mathematically unmixed to isolate discrete population of magnetic grains (Fig. 6). The deconvolution technique of Heslop et al. (2002) fits log-normally distributed populations of coercivities to high-resolution remanence acquisition (or demagnetization) data. Fig. 6 shows the results of unmixing on 18 different specimens selected from across all four sampling sites. The mean coercivity of each component is plotted against its corresponding dispersion parameter. Results from ARM and IRM

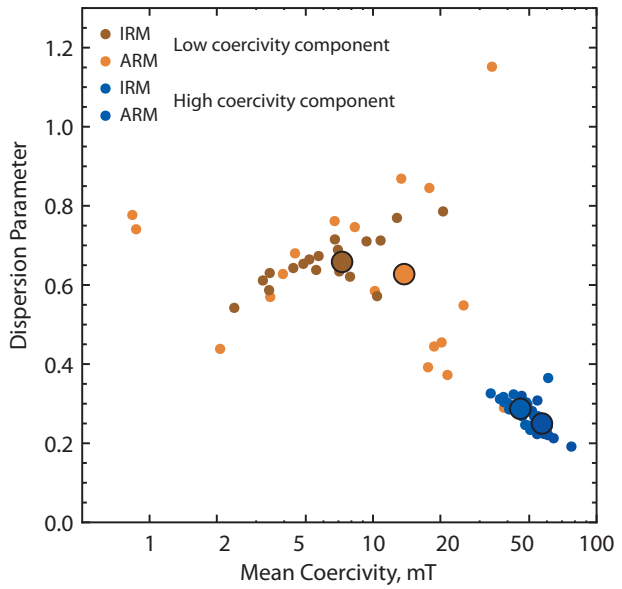


Fig. 6. Unmixing results from IRM and ARM acquisition experiments on 18 specimens. Mean values for the low and high coercivity components are shown using larger circles.

acquisition data agree well with one another (as well as with the FORC data), suggesting that these mathematically derived components are robust. IRM and ARM acquisition measurements can be modeled with good agreement using two coercivity components: a low coercivity component (LCC) and a high coercivity component (HCC). Mean coercivities for the HCC are tightly clustered near 50 mT with low dispersion parameters, suggesting a narrow grain size distribution of stable, single domain grains. On average, the HCC carries 91% of the IRM and 77% of the ARM. By contrast, the mean coercivities of the LCC are lower (~10 mT) and more scattered than the HCC with significantly higher dispersion parameters, suggesting a broader distribution of larger grain sizes. On average, the LCC carries 6% of the IRM and 19% of the ARM. Specimens that display evidence of elevated mineral alteration tend to have a larger portion of their ARM held by the LCC. The information gained from the magnetic unmixing and FORC results allow us to more carefully interpret the natural remanent magnetization of the Modipe Gabbro, by relying primarily on the remanence directions held by coercivities higher than 25 mT.

5. Paleomagnetism

NRM intensities range from 5.33×10^{-2} to 19.2 Am^{-1} with a median value of 1.04 Am^{-1} Alternating field demagnetization of ~6 cm^3 specimens displays a range of demagnetization behavior, with most specimens showing a unidirectional decay to the origin, while others exhibit up to three distinct components of magnetization (Fig. 7). The additional components in some samples are characterized by low coercivities and inconsistent directions, which are inferred to reflect weathering, lightning strikes, and/or multidomain magnetite. A portion of these low coercivity directions is consistent with the modern field direction at the site. Median destructive fields range from 5 to 47 mT, with a median value of 16 mT, which agrees well with the bulk coercivity values determined from major hysteresis loops. The strong-field thermomagnetic and susceptibility measurements described above suggest that this remanence is retained in nearly stoichiometric magnetite. The spectrum of coercivities shown in FORC diagrams (<1 to ~90 mT) indicates a mixed assemblage of magnetic domain states including large multidomain grains as well as nanometre-scale single domain grains. The characteristic remanent magnetization (ChRM) at all four sites shows reversed polarity (considering upward-directed magnetizations to be primary). Six to eight specimens per sampling locality yield well-clustered site mean directions (concentration parameter k = 31–248). Paleomagnetic data for these four sites and the comparable data from Evans and McElhinny are shown in Table 1 and illustrated in Fig. 7.

The mean declination (D) and inclination (I) for the Modipe Gabbro, as determined from the measurements in this study, are (D = 206.3°, I = 87.2°, α_{95} = 7.0°, k = 175, N = 4), which is statistically indistinguishable from the Modipe mean direction of Evans and McElhinny (1966) (D = 155°, I = 85°, α_{95} = 5.44°, k = 80, N = 10), which was determined using both alternating field and thermal demagnetization. Although modern least-squares fitting and great circle techniques were not applied to the demagnetization data collected by Evans and McElhinny (1966), the ChRM recorded by the Modipe Gabbro is sufficiently straightforward that their earlier site mean is considered to be reliable. When we combine the data from this study with that of Evans and McElhinny (1966), we arrive at a grand mean of D = 163.4°, I = 86.0°, α_{95} = 4.1°, k = 95, N = 14. If we exclude sites from Evans and McElhinny (1966) that had N < 4 or α_{95} > 15°, then we arrive at a grand mean of D = 164.6°, I = 86.1°, α_{95} = 5.0°, k = 84, N = 11.

The lack of any clear paleohorizontal indicator within the Modipe Gabbro has impaired the widespread use of the unit's paleopole in tectonic reconstructions. There is no evidence of compositional layering or mineral foliation at any of the sites visited in study, and measurements of magnetic fabrics (AMS and AARM) yielded anisotropies less than 1%. Without any bedding correction, the Modipe's ChRM indicates a paleolatitude of 82 ± 9°, which is relatively high when compared to other nearby units (the Derdepoort basalts, Wingate, 1998). However, the precise 2784.0 ± 1.0 Ma baddeleyite age described above suggests a link between the emplacement of the Modipe Gabbro and the eruption of the nearby 2782 ± 5 Ma Derdepoort basalts that outcrop along the Marico River ~20 km to the east (U–Pb SHRIMP on baddeleyite, Wingate, 1998). The mean attitude of these basalts is 125° strike and 26° SW dip, with local strikes ranging from 106° to 129°, and dips from 17° to 30° (Wingate, 1998). The mean strike is remarkably similar to the overall 126° trend of the chain of low-rising hills that mark the expression of the Modipe Gabbro. For this reason, we argue that the Modipe Gabbro is itself tilted along a similar strike to the Derdepoort basalts, and we apply a tilt correction of 11° around an axis of 125° in order to bring the paleolatitude of the Modipe Gabbro into agreement with that of the tilt-corrected Derdepoort basalts (Fig. 7). We have thereby calculated the paleolatitude of the Modipe Gabbro at 65.4 ± 8.6°. It should be noted that the quality of this tilt-correction is only as good as the quality of the paleomagnetic direction derived for the Derdepoort basalts (D = 222.7°, I = 76.6°, α_{95} = 10.4°) (Wingate, 1998). The importance of the precision of the Derdepoort paleomagnetic direction will be discussed in more detail below. It should also be noted that linking the paleomagnetic direction of the Modipe Gabbro with the Derdepoort basalts is based on the assumption that the two units are coeval within uncertainties and do not record significant variations in the geomagnetic field between their respective times of emplacement. We feel that given the structural and geochronological affinities between the Modipe Gabbro and the Derdepoort basalts, it can be assumed that they preserve a similar time-averaged record of the Earth's geomagnetic field.

6. Discussion

6.1. The magnetic carrier of the Modipe Gabbro

Petrographic and rock magnetic observations indicate that the Modipe Gabbro is variably altered, with increasing levels of

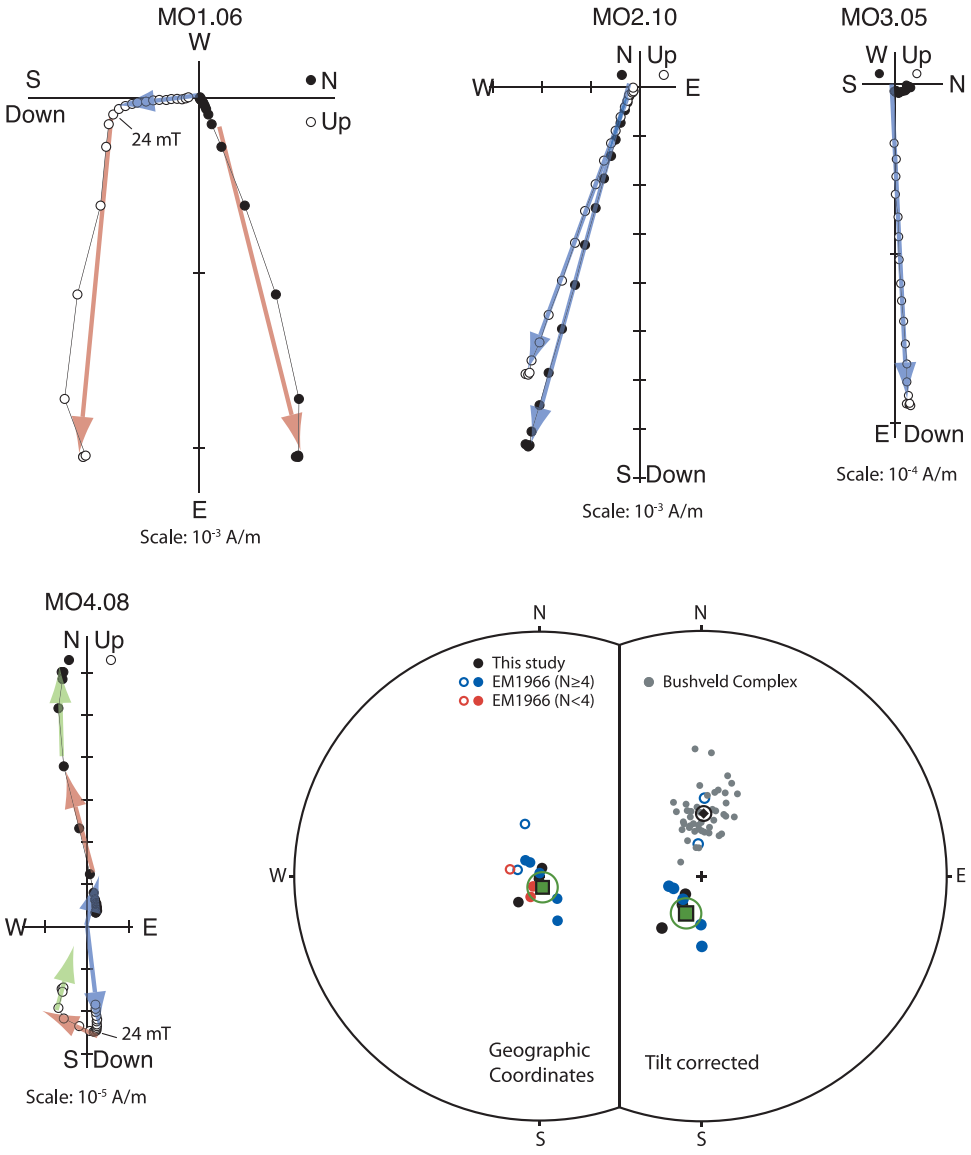


Fig. 7. Representative vector-component (Zijderveld) diagrams for the Modipe Gabbro acquired during alternating field demagnetization. The points used to define a ChRM are traced with a blue arrow, while secondary magnetizations are traced with a red or green arrow. Some specimens show univectorial decay toward the origin, while other display up to three components of magnetization. The equal area stereonets show the site mean directions for the Modipe Gabbro from this study (black circles), from Evans and McElhinny (1966) (blue circles). The grand mean direction for the Modipe Gabbro is shown with a green square. The red points represent sites from Evans and McElhinny (1966) with $N < 4$ and are not included in the calculation of the grand mean. The light gray circles in the tilt corrected data are the site mean directions of the eastern and western portions of the Main Zone of the Bushveld Complex (Hattingh, 1986a,b), showing that the remanence of the Modipe Gabbro is distinct from that of the Bushveld Complex. Solid (hollow) points fall in the lower (upper) hemisphere. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

alteration leading to higher concentrations of low coercivity magnetite. Much of this secondary magnetite was produced during the replacement of Fe–Mg-silicates, such as pyroxene and olivine. Despite this alteration, it is still possible to obtain convincing ChRMs related to the emplacement of the gabbro. Specimens with minimum levels of alteration display univectoral decay paths during alternating field demagnetization with coercivity distributions similar to the paleomagnetic recorders originally described by Evans and McElhinny (1966). Although we were unable to directly observe the exsolved pyroxene-hosted magnetite inclusions described by Evans et al. (1968) in any of our thin sections, there is clearly a high coercivity component within our samples that records a remanence identical to that reported in Evans and McElhinny (1966). Differences in paleomagnetic sampling strategies may offer a possible explanation for the apparent absence of inclusions in our samples. Evans and McElhinny (1966) used a specialized drill that was bolted directly into the outcrop prior to drilling. This arrangement provided a high degree of drilling stability, which allowed for the collection of a small number of especially long cores up to 60 cm long. By contrast, our samples were collected using a hand-held paleomagnetic drill, which allowed us to collect more samples per site, but at an average length of only 10 cm.

6.2. *Age of the Modipe Gabbro and its relationship to other igneous events*

The baddeleyite age of 2784.0 ± 1.0 Ma allows us to fit the emplacement of the Modipe Gabbro into a clearer geologic context than was previously possible using the existing geochronology (Supplemental Data File: Tables 1 and 2; Fig. 4). We note here that the previous data of McElhinny (1966), when corrected for modern radioactive decay constants, are wholly consistent with our results

Table 1
Magnetization directions of sites in the Modipe Complex, South Africa.

Site	Location		Characteristic remanent magnetization					Virtual geomagnetic pole				
	Lat. (°S)	Long. (°E)	*N*	*D* (°)	*I* (°)	α_{95} (°)	*k*	Paleolatitude (°)	Lat. (°S)	Long. (°E)	dp	dm
MO1	24.692	26.197	8 of 9	17.7	87.2	6.2	91	84.4 ± 12.2	19.4	28.0	12.3	12.4
MO2	24.680	26.217	6 of 10	218.9	78.9	13.6	31	68.6 ± 23.8	39.7	8.9	24.5	25.8
MO3	24.713	26.233	6 of 8	191.5	89.1	4.3	248	88.2 ± 8.5	26.5	25.8	8.6	8.6
MO4	24.699	26.197	8 of 9	156.4	86.6	6.5	84	83.2 ± 12.7	30.9	29.4	12.9	12.9
Mean of four site directions				206.3	87.2	7.0	175	84.4 ± 13.7	29.7	23.3	13.9	14.0
Data from Evans and McElhinny (1966)												
QA[a]	24.667	26.167	2	203.0	82.5	6.7	1400	75.2 ± 12.6	38.1	18.9	12.7	13.1
QB	24.683	26.183	4	325.5	84.5	8.5	120	79.1 ± 16.2	15.6	19.8	16.5	16.8
QC	24.650	26.150	4	286.0	−82.5	12.7	54	75.2 ± 23.3	27.8	42.2	24.2	24.8
QD	24.633	26.167	4	344.5	−72.0	7.1	167	57.0 ± 11.0	55.8	41.2	11	12.5
QF	24.683	26.183	4	3.5	89.0	3.6	653	88.0 ± 7.2	22.7	26.3	7.2	7.2
QJ	24.683	26.200	4	142.0	80.5	5.5	283	71.5 ± 10.1	38.6	40.7	10.2	10.6
QK	24.700	26.200	4	318.5	83.0	3.6	653	76.2 ± 6.9	14.1	16.8	6.9	7
QL[a]	24.683	26.183	3	211.5	86.0	3.9	994	82.3 ± 7.7	31.4	21.3	7.7	7.7
QM[a]	24.650	26.150	3	283.0	−80.0	18.7	45	70.6 ± 32.3	27.5	47.6	34.3	35.8
BO	24.667	26.167	4	158.0	74.0	1.7	2900	60.2 ± 2.8	51.4	43.5	2.8	3.1
Grand mean of fourteen site directions				163.4	86.0	4.1	95	82.1 ± 8.0	32.3	28.9	8.1	8.1
Grand mean of eleven site directions				164.6	86.1	5.0	84	82.2 ± 8.7	32.1	28.8	9.9	9.9
After bedding correction of 125.5° 10.6°				202.3	76.6	5.0	84	64.5 ± 8.6	47.6	12.4	8.6	9.3

N is the number of core samples; *D* is the declination of the magnetization direction, measured eastward from north; *I* is the inclination of the magnetization direction, downward positive, upwards negative; α_{95} is the semi-angle of cone of 95% confidence; *k* is the estimate of the precision parameter *k*.

[a] Site was excluded because $N < 4$ or $\alpha_{95} > 15$.

albeit with much larger uncertainties. The average K–Ar age of pyroxenes (2758 ± 126 Ma) reported by McElhinny (1966) is indistinguishable from our $^{207}Pb/^{206}Pb$ baddeleyite age, consistent with quantitative retention of radiogenic argon in the pyroxene despite a reheating event that reset the average plagioclase K–Ar age to 1961 ± 52 Ma. The apparent retentivity of argon in the pyroxene is consistent with the results of Cassata et al. (2011) which indicate closure temperatures of 700–800 °C at cooling rates of 10 °C/Ma. Combined with diffusion kinetics for argon in plagioclase (Cassata et al., 2009), the K–Ar results delineate a range of possible thermal histories for the Modipe Gabbro resulting from reheating effects of the Bushveld Complex, as shown in Supplementary Data File Fig. 4.

The Modipe Gabbro can now be temporally linked to other large igneous events occurring within the Kaapvaal craton, including the Gaborone Granite Suite at 2783.1 ± 2.0 Ma (U–Pb TIMS, multi-crystal zircon, Moore et al., 1993), the Kanye Rhyolitic Volcanics at 2784.7 ± 1.7 Ma (U–Pb TIMS, multi-crystal zircon, Moore et al., 1993), and the Derdepoort Basalts at 2782 ± 5 Ma (U–Pb SHRIMP on baddeleyite, Wingate, 1998). While direct comparison of geochronological data derived from these different U–Pb radio-isotopic analytical methodologies may be problematic, there is now ample evidence for significant magmatism within the Kaapvaal craton at ~2784 Ma with a minimum areal extent of 6000 km^2. Importantly, this igneous activity is represented by well-preserved felsic and mafic intrusive and extrusive rocks, and as such is one of the oldest fully differentiated magmatic events available for study in the geologic record.

The driving force for this igneous event is thought to be related to the Archean Limpopo orogeny, attributed to a continental collision along the northern edge of the Kaapvaal craton (Burke et al., 1986). The emplacement of mantle-derived magmas near the base of the thickened crust initiated melting in the overlying rock, producing the magmas that gave rise to the cogenetic mafic and felsic assemblage of intrusive (e.g., the Gaborone Granite and Modipe Gabbro) and extrusive (e.g., Kanye Volcanics and Derdepoort Basalts) igneous systems (Moore et al., 1993). The clustering of U–Pb ages from each of these systems implies that their formation was rapid, perhaps on the order of 1–3 million years.

6.3. Vaalbara hypothesis

The Vaalbara hypothesis (Cheney, 1996; Zegers et al., 1998) is based on the similarities found in lithostratigraphy and inferred tectonic histories of the Pilbara craton of western Australia and the Kaapvaal craton of southern Africa. These Archean cratons, the best-preserved vestiges of Archean crust still in existence, are proposed to have formed a super-craton named Vaalbara during the period 2.7–2.1 Ga based on the sequence stratigraphy of their cover sediments (Cheney, 1996), and paleomagnetism and alignment of tectonic lineaments (Zegers et al., 1998; Strik et al., 2003; de Kock et al., 2009), in which the Pilbara craton is located immediately to the northwest of the Kaapvaal craton (in today's Kaapvaal reference frame).

The comparison of well-dated paleomagnetic poles is the only quantitative means of constraining relative plate positions through time. Our data dramatically improve the geochronological constraints on the paleomagnetic data of Evans and McElhinny (1966), thereby further restricting the possible location of the Kaapvaal craton in the late Archean. Our 2784.0 ± 1.0 Ma virtual geomagnetic pole (VGP) of 47.6° S, 12.4° E, dp = 8.6°, dm = 9.3° is considered to be primary for several reasons. First, the ChRMs of the northernmost samples located on the Botswana side of the border are antipodal to those found on the South African side of the border. As above, we consider those with a positive inclination to be "N", and those with a negative inclination to be "R". The comparison of the "N" samples ($D = 182.2°$, $I = 87.6°$, $N = 9$, $k = 102$, $\alpha_{95} = 5.1°$) and "R" samples ($D = 328.1°$, $I = -78.6°$, $N = 2$, $k = 55$, $\alpha_{95} = 34.2°$) results in a positive reversal test using the "Simulation" method of McFadden and McElhinny (1990) to account for the low *N* value of the "R" population. This test yields a positive B classification, as the angle between the mean of the normal and the mean of the reversed polarity sets lies between 5° and 10° ($\gamma_o = 9.5°$, $\gamma_c = 22.2°$). Second, the unblocking temperatures of the Modipe Gabbro samples are ca. 580 °C, and in order to reset their TRM, local temperatures would have had to equal or exceed this value. While there is evidence for widespread hydrothermal alteration and/or low-grade metamorphism up to greenschist facies (see above), temperatures do not seem to have exceeded 500 °C in these rocks. Third, the Modipe

ChRM is very different from that of the nearby ca. 2050 Ma Bushveld Complex (Evans et al., 2002; Fig. 4), suggesting that the emplacement of the large layered igneous intrusion did not overprint the Modipe Gabbro. Fourth, the Modipe Gabbro is now shown to be temporally associated with the Derdepoort lavas, which because of the tilt-correction used here, have a paleopole (Wingate, 1998) that is statistically indistinguishable from that of the Modipe Gabbro (though less precise) and that passes an intraformational conglomerate test and is therefore itself convincingly primary. For these reasons, we conclude that the Modipe paleopole presented here is primary and suitable for use in comparison with other cratons.

The only reliable paleopole with which to compare this pole is within the Pilbara craton, specifically the rocks of the lower Fortescue Group of northwestern Australia. These are primarily the basalts of the Nullagine Supersequence, the oldest of which, known as "Package 1" by Strik et al. (2003), are the Mount Roe Basalts and the affiliated Black Range Suite of mafic intrusive rocks (Blake, 2001). The Black Range Suite has been dated using the SHRIMP method ($^{207}Pb/^{206}Pb$ age on baddeleyite (Wingate, 1999)), with four samples dated between 2768.0 ± 2.7 Ma and 2774.7 ± 2.2 Ma. The paleomagnetism of this package (Strik et al., 2003; Schmidt and Embleton, 1985) yields a paleopole of 40.8° S, 159.8° E, N = 13, α_{95} = 3.7°, which passes a fold test and conglomerate test, though the timing of these geological features is not well-established but is apparently Archean. While it is not always appropriate to directly compare geochronological data from ID-TIMS studies such as this one and ion microprobe studies such as Wingate (1999), taken at face value the Pilbara and Kaapvaal poles are ca. 10 Ma apart in age, and relative drift in that time is unknown. However, given these limitations with respect to their direct comparisons, these paleomagnetic datasets represent the oldest reliable opportunity presently available to reconstruct the relative positions of Archean cratons.

Wingate (1998) determined the paleolatitude of the Derdepoort basalts to be 64.5 ± 17.5°, and concluded that the Kaapvaal craton was likely not attached to the Pilbara craton based on the nearest approach permitted by the Mount Roe basalt (part of the Fortescue Package 1), which he determined to have a paleolatitude of 34.3 ± 6.4°. These paleolatitudes, within their uncertainties, permit a nearest approach of 6.3° (a distance of over 700 km) and therefore a wide gap between the two cratons and an invalid Vaalbara hypothesis. However, Strik et al. (2003) refined the Pilbara paleolatitude to include all of the units making up Fortescue Package 1, and their new value of 51.1 ± 3.9° indicated significant overlap with the Derdepoort paleolatitude and permitted the joining of the Kaapvaal and Pilbara cratons. de Kock et al. (2009) used "pole pairs" (essentially short segments of apparent polar wander paths) of the above paleopoles and the next most recent reliable paleopoles at ca. 2.7 Ga to constrain relative paleolongitude between the two cratons (assuming a shared drift history). This resulted in their fit, which also places Pilbara in close proximity to Kaapvaal's northwest corner (in modern Kaapvaal's reference frame). All of the above reconstructions and constraints have been hampered by the relatively large uncertainty on the 2.78 Ga Kaapvaal pole from the Derdepoort basalts, allowing a significant amount of leeway in the alignment of the cratons.

When the paleomagnetic data from the Modipe Gabbro are taken at face value (paleolatitude 64.5 ± 8.6°), the "nearest approach" analysis results in a gap between the Kaapvaal and Pilbara sites of 1.8°, or ~200 km (Fig. 8). Paleolongitude is unconstrained in this analysis and the sampling sites could be up to 180° apart if the alternate polarity is used. However, as the precise contemporary boundaries of the cratons at 2.78 Ga are not well-established, and the sites sampled for paleomagnetism (and thus the locations to which the calculated paleolatitudes actually apply) are not at the closest edge of the cratons, it is likely that

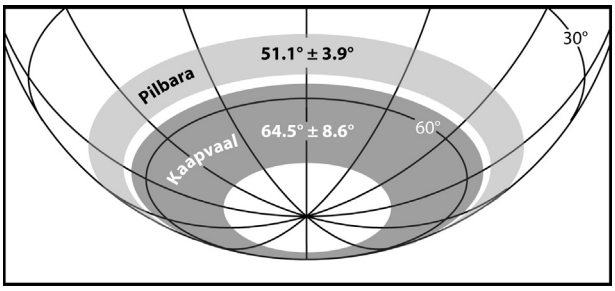


Fig. 8. Paleolatitude reconstructions for the Pilbara (light gray) after Strik et al. (2003), and the Kaapvaal (dark gray) at ~2.78 Ga. The gray bands indicate the paleolatitudes of the sampling sites and their 95% confidence limits. The possible range of Pilbara paleolatitudes does not overlap with the possible range of Kaapvaal paleolatitudes from the tilt-corrected Modipe Gabbro. However, the possible paleolatitude ranges do overlap when the α_{95} value of the Derdepoort basalts is applied to the Modipe Gabbro data.

the cratons' edges extend beyond the uncertainty limits of Fig. 8 and a close fit within the uncertainties of the datasets is still possible. An important caveat to this analysis is that we use the error on the Modipe Gabbro site direction (α_{95} = 5.0°) to establish the paleolatitude error range shown in Fig. 8. Yet as mentioned earlier, the tilt-corrected Modipe direction is only as good as the Derdepoort site direction, which has a significantly larger α_{95} of 10.4° (Wingate, 1998). When Wingate's (1998) α_{95} value is used, the error on the range of possible Modipe Gabbro paleolatitudes increases by more than a factor of two, overlapping significantly with that of the Pilbara sites, or alternatively, permitting a significantly greater spacing between the cratons.. Yet another consideration is that the pole we present here is a VGP rather than a mean pole (incorporating multiple VGPs), as the paleomagnetic data come from a single cooling unit. However, the cooling rate for large intrusive bodies such as the Modipe Gabbro is typically on the order of hundreds to thousands of years, and as different parts of the rock unit cool below the 580 °C unblocking temperature, the TRM records the field direction at that time. Our collection and measurement of several samples from different locations across the Modipe Gabbro should therefore represent an average of secular variation over the entire duration of cooling, and thus the VGP can be treated as a robust paleopole. Therefore, while a Vaalbara reconstruction at 2.78 Ga is possible given the uncertainties of both the Kaapvaal and Pilbara paleomagnetic data, as recently reviewed by de Kock et al. (2009), a generous interpretation of the various uncertainties is required.

7. Summary and conclusions

Despite its great antiquity, the Modipe Gabbro offers a clear view into the late Archean. Although rock magnetic and petrographic observations show the Modipe Gabbro to be variably altered, perhaps to lower greenschist facies, the Modipe Gabbro's stable and robust remanence allow the derivation of convincing ChRMs related to the original emplacement and cooling of the gabbro. The Modipe Gabbro's emplacement age of 2784.0 ± 1.0 Ma indicates that it was part of a larger, rapidly emplaced igneous province formed within the Kaapvaal craton at ~2785–2783 Ma. This age allows us to apply a tilt correction to the mean paleomagnetic direction of the Modipe Gabbro, which brings its paleolatitude into agreement with that of the nearby Derdepoort basalts (Wingate, 1998). It is important to emphasize that this tilt-correction is based not only on the similar emplacement ages of the two units, but also on field observations that show that the strike of the Modipe Gabbro is consistent with the mean strike of the bedding of the Derdepoort basalts. The combination of a highly precise emplacement age and newly refined paleomagnetic analyses allow us to include the magnetization of the Modipe Gabbro in discussions of the Vaalbara

hypothesis. Our paleomagnetic data show that the Kaapvaal craton was positioned at latitudes that may overlap the possible position of the Pilbara craton within uncertainties, in modest support of the Vaalbara hypothesis.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.precamres.2013.10.002.

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